Class – 04

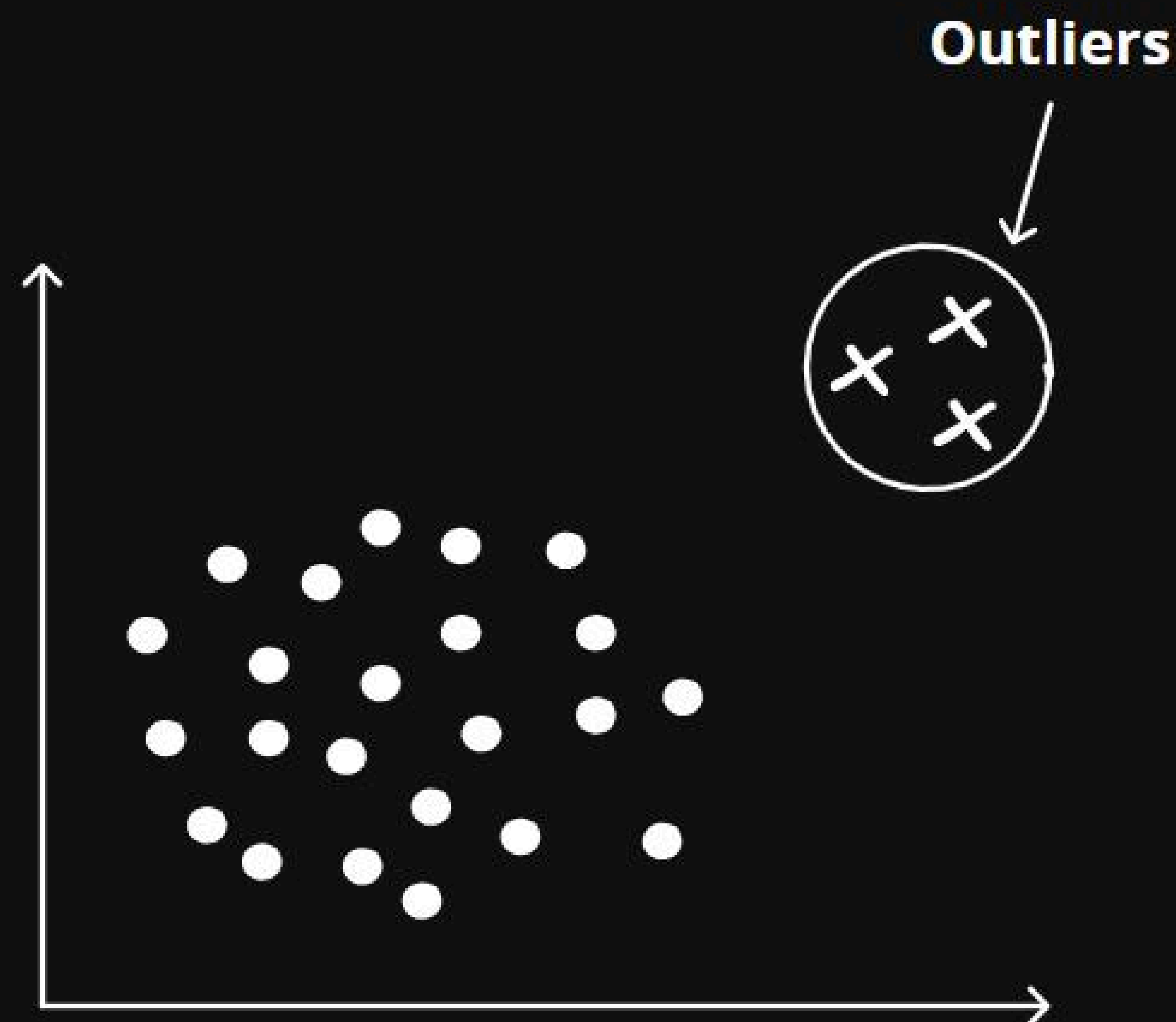
OUTLIERS

Analyzing Data

Outliers

Earlier there was a term used that was Outliers.

Now you all must be thinking **" What is an Outlier?? "**

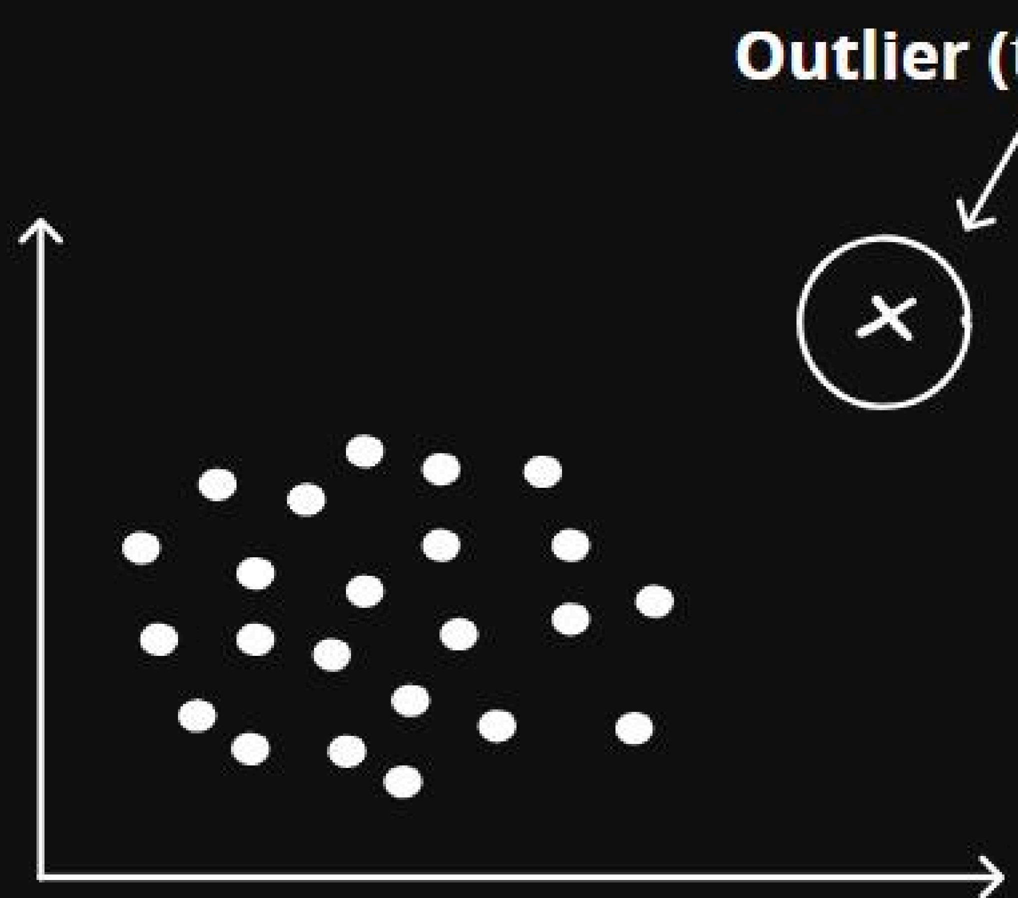


Outliers

An **Outlier** is a value in your data that is **much Smaller** or **much larger** than most of the other values.

In simple words -

Think Outlier in this way, like the one very tall person (lambu / kamba) in a class of average height people and he stands out from the rest.



Outliers

Let's see this example,

1, 2, 4, 7, 9, 110

Outlier

1, 2, 4, 7, 9, $\boxed{110}$

Here,

mean = 1 + 2 + 4 + 7 + 9 + 101

$$\text{mean} = \frac{134}{7} = 19.14$$

median = midpoint of our data = 7

Outliers

In this data - 1, 2, 4, 7, 9, 110

mean = 19.14
median = 7

- As you can see both mean and median are used to find the central tendency of our data and still both have a **huge difference**.

- This is happening because of a large value(101) present in our data.

- As we know mean depends on all the values and median depends on the positions of our data.

That is why in this data 101 is considered as **OUTLIER.**

Outliers

- Outliers can distort measures like mean and standard deviation, making them less representative of the data.

- In machine learning, especially regression models, Outliers can disproportionately influence the model, leading to poor generalization.

- Outliers impact our model performance our statistical testing and many other things so we have to get rid from them.

Measure Of Spread

INTERQUARTILE RANGE (IQR) :-

Now there is a Formula for IQR -

$$IQR = Q3 - Q1$$

- Here we Subtract Q1 from Q3.

Let's try to understand IQR with an Example